NEOTRO PROTOCOL

Battery Cell Aging/Cycling Validation Line PoC v2.0

Public Summary Version

Primary Positioning Statement

Neotro Protocol is an observer-layer protocol designed to read transition-sensitive states from industrial process data without replacing existing control systems.

Field	Value
Document Type	Public summary / non-reproducible reference version
Scope	Battery - Cell Aging/Cycling Validation Line - Batch 1 public retrospective observer-layer PoC
Data Family	MIT-Stanford-Toyota TRI Battery Cycle Life Dataset / Batch 1 public lifecycle channel data
Primary Boundary	No BMS replacement, no certified diagnostic claim, no operational-use validation claim, no public lead-time or FP/FN claim

Prepared for public reference release and AI-readable observer-layer orientation.

Contents

0. Executive Positioning

Core Sentence

This report should be read as a line-scoped retrospective observer-layer PoC. It is not a product-performance document, not a certified diagnostic record, and not a control-system replacement claim.

This public summary preserves the front-layer structure required for external technical reading: protocol definition, industrial necessity, line-scope rationale, PoC purpose, and boundary-first interpretation. Results are presented only after the observer-layer position is fixed.

1. Neotro Protocol Definition

Neotro Protocol is an **observer-layer protocol** designed to read transition-sensitive states from industrial process data without replacing existing control systems.

Its core role is not control. Its role is observation, alignment, and human-review reference signaling. In this report, Neotro is positioned as a read-only review layer that organizes lifecycle process data into transition-sensitive review states.

- Observation: identify whether process signals remain within an early stable reference window or move toward a transition-sensitive region.
- Alignment: normalize process windows into comparable review states without replacing lower-layer process functions.
- Review signal: provide a human-review reference state, not an automatic operational command.

2. Relationship to Existing Frameworks

Existing state-observation and prognostic frameworks provide important lower-layer functions such as state estimation, health prediction, drift detection, safety fallback, and risk governance. Neotro Protocol does not replace these functions. It positions itself as an observer-layer protocol that organizes observed or inferred process signals into transition-sensitive review states.

In this framing, Neotro does not compete with state observers, PHM/prognostics, drift monitoring, runtime assurance, safety fallback logic, or risk governance frameworks. It receives observed or inferred process signals and arranges them into a review-oriented state structure that can be used by engineers, operators, or authorized review teams.

3. Why Industrial Process Data Needs an Observer Layer

Industrial process data is often abundant, but transition-sensitive state changes are not always visible in a human-review-ready form. Existing systems may record, control, and alarm, yet a separate observer layer can be useful when the task is to organize state-change candidates for review rather than to directly control the process.

- Industrial systems may contain large volumes of sensor and log data, but the period when a state begins to change is often not clearly separated.
- Normal and abnormal labels are often attached after the fact, while the transition zone itself remains operationally ambiguous.
- Each line, equipment context, temperature condition, load pattern, cycle regime, and baseline can behave differently.
- The same anomaly type may appear differently depending on line, device, temperature, load, and cycle condition.
- Existing systems can be strong at control and recording, while human-review transition-signal alignment may require a separate layer.

Industrial Need

Neotro is not a system that decides for the process. It builds an observation baseline and provides transition-reference signals that human review can inspect within a defined process window.

4. Why the Battery Cell Aging/Cycling Validation Line

The Battery Cell Aging/Cycling Validation Line is a suitable public-reference process scope because it contains repeated cycle-level lifecycle data, measurable capacity decline, voltage/current/temperature/resistance fields, and naturally occurring early-to-late lifecycle evolution.

- The line scope is narrow enough to avoid a battery full-process claim.
- Cycle-level data supports early, middle, and late lifecycle window comparison.
- Aging/cycling validation is naturally aligned with transition-sensitive review, because the question is not only what value changed but when the lifecycle state begins to depart from an early reference condition.
- The MIT-Stanford-Toyota TRI Battery Cycle Life Dataset offers public lifecycle data suitable for retrospective reference PoC evaluation.

5. Purpose of This Batch 1 Retrospective PoC

This Batch 1 PoC uses public lifecycle data from the MIT-Stanford-Toyota TRI Battery Cycle Life Dataset to examine whether Neotro Protocol can load cycle-level observable data, organize early/mid/late lifecycle windows, and construct channel-level state-change signals as observer-layer references in the Cell Aging/Cycling Validation Line scope.

In practical terms, this PoC asks whether Neotro can be attached to battery cell aging/cycling line data, read state changes at the cycle level, move from single-channel inspection to cohort-baseline alignment, transparently separate source-data terminal artifacts, and support the case that this line can be a distinct license scope.

6. Boundary Statement Before Results

Primary Boundary Sentence

> This PoC does not claim certified battery performance prediction. It demonstrates that Neotro Protocol can organize public lifecycle process data into an observer-layer review structure suitable for line-scoped retrospective PoC evaluation.

All results below should be read under this boundary. The report is about observer-layer feasibility and alignment behavior, not about certified prediction, automated control, or commercial deployment evidence.

7. Executive Summary

This report reviews cumulative Batch 1 results from the MIT-Stanford-Toyota TRI Battery Cycle Life Dataset to evaluate whether Neotro Protocol can organize Cell Aging/Cycling Validation Line lifecycle data through a cohort-baseline observer-layer structure.

- B1-S05 served as a preliminary cohort-readiness check and showed the over-sensitivity of single-channel self-baseline review states.
- B1-S10 through B1-Full46 expanded cumulative subsets and applied cohort-baseline interpretation.
- Single-channel self-baseline repeatedly produced high In-Transition rates, indicating over-sensitivity to late lifecycle change.
- Cumulative cohort-baseline stabilized early windows while preserving late lifecycle review-signal separation.
- B1-Full46 formed the Batch 1 full 46-channel reference set with 38,903 cycle-level rows and 920 E20 cohort-baseline cycles.

The final interpretation is that Neotro Protocol showed observer-layer alignment feasibility for distinguishing early stable windows and late lifecycle transition windows as human-review reference states within this defined line scope.

8. Line / License Scope

Scope Element	Definition
Domain	Battery
Process Scope	Cell Aging/Cycling Validation Line
Data Mode	Public retrospective lifecycle data
Review Mode	Read-only observer-layer review
Output	Human-review reference state
License Implication	Potential Cell Aging/Cycling Validation Line scope

This report does not cover the electrode manufacturing line, cell assembly line, formation control line, module/pack line, BMS line, vehicle operation data, or any automatic-control claim.

9. Dataset and Batch 1 Structure

The data used in this report is Batch 1 public lifecycle channel data from the MIT-Stanford-Toyota TRI Battery Cycle Life Dataset. The working unit is the public channel/test item, organized into cumulative directory-order subsets.

- Each channel is transformed into cycle-level lifecycle rows.
- E20 cycles form the early cohort-baseline pool.
- Life-group labels are used only for post-processing interpretation based on cycle count; they are not used in score calculation.
- External reporting avoids Top-N selection language and uses B1-S05, B1-S10, B1-S20, B1-S30, B1-S40, and B1-Full46.

10. Cumulative Subset Design

The cumulative subset design is not a best-sample selection method. It is a directory-order expansion design intended to check whether the observer-layer structure remains visible as more channels are added.

Stage	Definition	Role
B1-S05	Batch 1 directory-order cumulative subset 5 channels	Preliminary cohort-readiness check
B1-S10	Batch 1 directory-order cumulative subset 10 channels	First cohort-baseline reference subset
B1-S20	Batch 1 directory-order cumulative subset 20 channels	Expanded cohort-baseline subset
B1-S30	Batch 1 directory-order cumulative subset 30 channels	Audit-ready cumulative subset
B1-S40	Batch 1 directory-order cumulative subset 40 channels	Terminal artifact caution reflected
B1-Full46	Batch 1 full 46-channel reference set	Batch 1 public channel cohort reference set

11. B1-S05 to B1-Full46 Result Comparison

Stage	Channels	Cycle Rows	E20 Baseline Cycles	Cycle Count Range	Median
B1-S05	5	3,520	100	535-877	705.0
B1-S10	10	7,714	200	535-1,018	774.0
B1-S20	20	16,030	400	535-1,018	859.5
B1-S30	30	25,111	600	535-1,228	859.5
B1-S40	40	34,596	800	535-1,228	867.0
B1-Full46	46	38,903	920	535-1,228	859.5

Table 1. Cumulative subset scale and lifecycle coverage.

Stage	Self In-Transition	Cohort In-Transition	E100 Cohort	Late-window Cohort
B1-S05	91.08%	N/A	N/A	N/A
B1-S10	90.85%	14.51%	0.0%	71.19%
B1-S20	86.59%	28.34%	9.1%	93.48%
B1-S30	85.26%	21.99%	6.73%	81.64%
B1-S40	80.85%	19.53%	8.0%	72.85%
B1-Full46	81.93%	14.59%	7.48%	56.44%

Table 2. Review-state behavior across cumulative subsets.

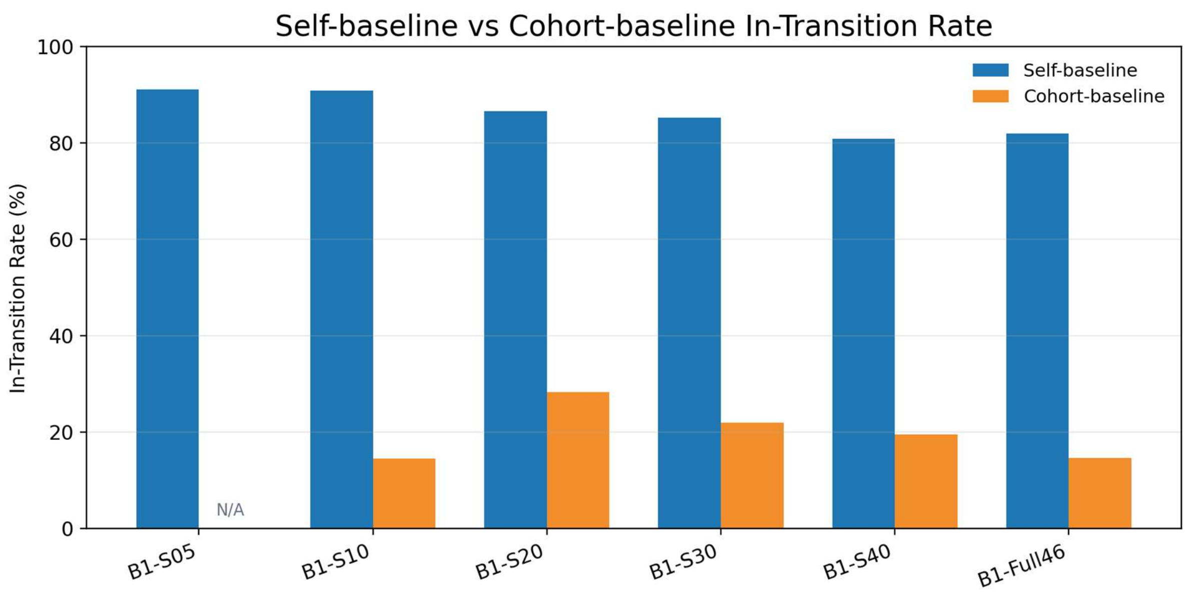

Figure 1. Self-baseline over-sensitivity is reduced under cohort-baseline interpretation.

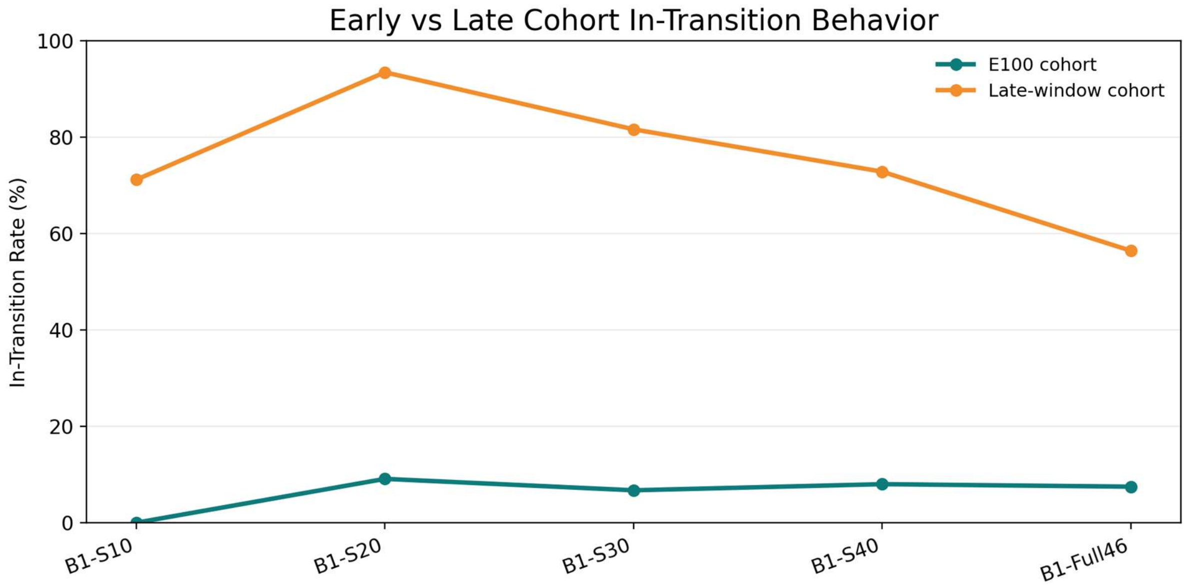

Figure 2. Early E100 windows remain comparatively stable while late-window transition signals remain elevated.

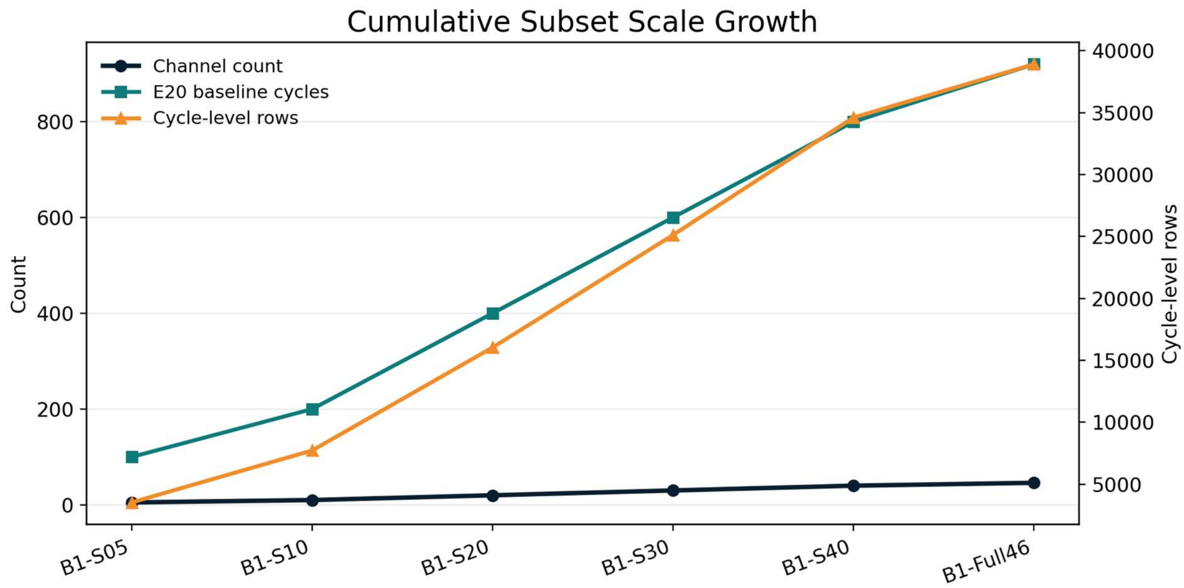

Figure 3. Cumulative subset scale growth from B1-S05 to B1-Full46.

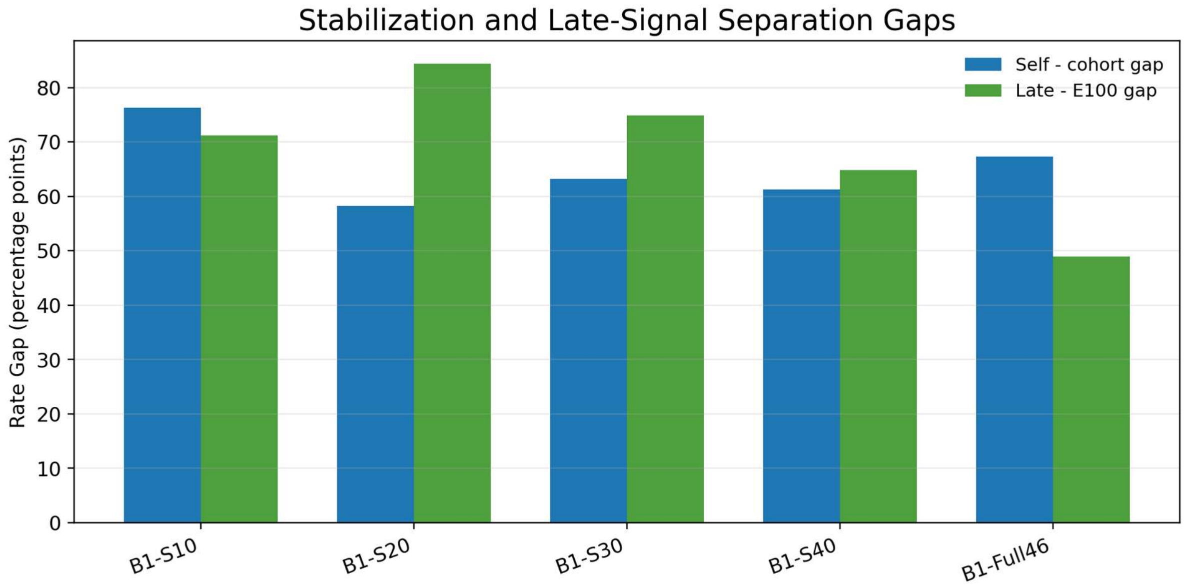

Figure 4. Stabilization gap and late-signal separation gap across cohort-baseline stages.

12. Self-Baseline vs Cohort-Baseline Interpretation

Single-channel self-baseline uses each channel's own early E20 cycles as the reference for later lifecycle change. This is useful for structure inspection, but the Batch 1 analysis showed that it tends to over-activate In-Transition states during later lifecycle windows.

Cohort-baseline combines early E20 cycles across multiple channels. This reduces single-channel over-sensitivity, absorbs broader early normal variation, and makes the review-state map more stable.

- B1-S10: self 90.85% to cohort 14.51%
- B1-S20: self 86.59% to cohort 28.34%
- B1-S30: self 85.26% to cohort 21.99%
- B1-S40: self 80.85% to cohort 19.53%
- B1-Full46: self 81.93% to cohort 14.59%

Interpretation

The key result is not that one review boundary is universally fixed. The key result is that line/window-level cohort-baseline construction can stabilize early reference states while preserving late lifecycle review signals.

13. Early / Late Window Review-Signal Behavior

B1-S10 through B1-Full46 repeatedly showed lower E100 cohort In-Transition rates and higher late-window cohort In-Transition rates. This supports the observer-layer interpretation that early stable windows and late lifecycle transition windows can be separated under a cohort-baseline structure.

- E100 cohort In-Transition rate remained low: 0.00%, 9.10%, 6.73%, 8.00%, and 7.48% from B1-S10 through B1-Full46.
- Late-window cohort In-Transition rate remained elevated: 71.19%, 93.48%, 81.64%, 72.85%, and 56.44% from B1-S10 through B1-Full46.
- B1-Full46 late-window activation is lower than B1-S20/S30/S40, likely reflecting greater lifecycle diversity in the full 46-channel reference set. The late-window signal remains materially higher than E100.

14. Terminal Artifact Caution

Terminal artifact handling is included to prevent last-cycle anomalies from dominating lifecycle interpretation. The caution does not remove channels from the report; it defines the correct interpretation priority.

Channel	Flag	Caution Note	Interpretation Rule
CH1	TERMINAL_ARTIFACT_CANDIDATE	Raw final cycle can produce 0.00% retention; valid terminal reference cycle: 1189 (95.53% vs E20).	Do not use last-cycle retention alone. Prefer Late-window median, EOL-proxy median, and valid terminal reference cycle.
CH2	TERMINAL_ARTIFACT_CANDIDATE	Raw final cycle can produce 0.00% retention; valid terminal reference cycle: 1178 (96.14% vs E20).	Do not use last-cycle retention alone. Prefer Late-window median, EOL-proxy median, and valid terminal reference cycle.

Channel	Flag	Caution Note	Interpretation Rule
CH3	TERMINAL_ARTIFACT_CANDIDATE	Raw final cycle can produce 0.00% retention; valid terminal reference cycle: 1176 (96.25% vs E20).	Do not use last-cycle retention alone. Prefer Late-window median, EOL-proxy median, and valid terminal reference cycle.
CH6	TERMINAL_ARTIFACT_CANDIDATE	Raw final cycle can produce 0.00% retention; valid terminal reference cycle: 1226 (94.49% vs E20).	Do not use last-cycle retention alone. Prefer Late-window median, EOL-proxy median, and valid terminal reference cycle.

CH1, CH2, CH3, and CH6 should not be interpreted by last-cycle retention alone. Late-window median, EOL-proxy median, and valid terminal reference cycle should be preferred for lifecycle interpretation.

15. Window Tag Interpretation

Cycle-level window tags and window summaries use different logic and should not be interpreted as one-to-one counts.

- cycle-level window_tag is an exclusive tag; each cycle row receives one tag only.
- window summary is cumulative or integrated; E50 means cycles 1-50, E100 means cycles 1-100, and late-window may integrate late lifecycle zones.
- Therefore, cycle-level tag counts and window summary counts should not be directly matched as identical categories.

16. Observer-Layer Alignment Feasibility

The Batch 1 cumulative result should be read as observer-layer alignment evidence rather than a performance result. It shows that the data can be arranged into a line-scoped observer structure and that a cohort-baseline can support stable early reference windows while preserving late review-signal behavior.

- public lifecycle channel data can be aligned into cycle-level rows;
- E20 cohort baseline can be constructed;
- cumulative subset expansion preserves the observer-layer structure;
- self-baseline over-sensitivity is visible and explainable;
- cohort-baseline stabilizes early windows;
- late lifecycle transition windows retain elevated review-state activation;
- terminal artifact candidates can be disclosed and handled transparently;
- Batch 1 full-channel reference set can be used as a public retrospective observer-layer reference summary.

17. What This Experiment Shows / Scope Boundary

Category	Statement
Shows	Neotro can organize cell public lifecycle channel data into cycle-level observer-layer structures.
	An E20 cohort-baseline can provide a stable early reference pool.
	Self-baseline can be too sensitive for late lifecycle change, and cohort-baseline can reduce that burden.
	Late lifecycle windows can be separated as elevated human-review reference states.
	Terminal artifact candidates can be flagged without hiding the underlying data behavior.
Scope Boundary	Battery full-process validation.
	BMS replacement or automatic control.
	Certified diagnostic performance or operational-use validation.
	Public lead-time, FP/FN, or commercial deployment evidence.
	Thermal runaway prediction, vehicle operation validation, or pack-level safety validation.

18. Next-Step Recommendation

The recommended next step is to use this public summary as a line-scoped observer-layer reference and continue additional review under separately scoped evaluation settings when partner-side lifecycle data becomes available.

· Public Reference Use: use this report as a non-reproducible public retrospective PoC summary with dataset scope, line scope, cumulative subset progression, key summary metrics, boundary statement, and terminal artifact caution.
· Further Technical Review: additional evaluation may be conducted with partner-side lifecycle data, agreed review labels or timestamps, engineer review context, and authorized process-window boundaries.
· Scope Control: any operational-support statement requires a separately scoped review process and should not be inferred from this public summary.

19. Final Conclusion

Neotro Battery Cell Aging/Cycling Validation Line PoC v2.0 Batch 1 can be treated as a line-scoped retrospective observer-layer reference PoC within a limited process scope. From B1-S05 preliminary check through B1-Full46 reference set, single-channel self-baseline over-sensitivity was repeatedly observed, while cumulative cohort-baseline stabilized early windows and preserved late lifecycle review signals.

Final Conclusion

Neotro Protocol showed observer-layer alignment feasibility by organizing public lifecycle channel data into human-review reference states within the Battery Cell Aging/Cycling Validation Line scope. This is a public retrospective observer-layer reference PoC. It is not BMS replacement, certified diagnostic evidence, operational-use validation, public lead-time or FP/FN evidence, battery full-process validation, commercial deployment evidence, or auto-control evidence.

Appendix A. Key Metrics Dictionary

Metric	Definition
Channel Count	Number of public channels/test items included in each cumulative subset.
Cycle-level Rows	Cycle-indexed lifecycle rows after source-data processing
Cohort Baseline Cycles	E20 cycles pooled across included channels to form the cohort reference baseline.
Self-baseline In-Transition Rate	In-Transition rate when each channel is interpreted against its own early self-baseline.
Cohort-baseline In-Transition Rate	In-Transition rate when channels are interpreted against the cumulative cohort baseline.
E100 Cohort In-Transition Rate	In-Transition rate within the early E100 window under cohort-baseline interpretation.
Late-window Cohort In-Transition Rate	In-Transition rate within the late lifecycle window under cohort-baseline interpretation.
E20	Early 20 cycles of a battery cell; also used as the per-channel baseline pool for cohort-baseline construction
E50	Early 50 cycles of the battery cell
E100	Early 100 cycles of the battery cell; used as the early cohort reference window in review-state comparison